

InsydeH2O® [AMD_V3000_Fp7MultiBoard_Rev5.4] Release Note

Contents

Tag Number	Date
Tag# 05.46.50.0022	2025/1/2

1. System Configuration

Hardware Configuration

CRB Name:	Fox
Platform Name:	Ryzen Embedded V3000
CPU:	FP7r2

Firmware Configuration

Update	Firmware Configuration Items	Current Ver. in BIOS	Latest Ver. by Chipset Vendor	Note
☒	Insyde Kernel Version	05.46.50	05.46.50	
☒	Reference Code Version	EmbeddedPI-FP7r2 1.0.0.C	EmbeddedPI-FP7r2 1.0.0.C	
☒	PSP	0.28.A8.70	0.28.A8.70	
☒	SMU(B0)	6.69.29.0	6.69.29.0	
	Ucode(B0)	0A404002	0A404002	
☒	Ucode(B1)	0A40410A	0A40410A	
☒	SmmSupervisor	3.011X	3.011X	
☒	CPM	v065	v065	

Driver List

Update	O.S.	Driver	Current Ver. in BIOS	Latest Ver. by Chipset Vendor	Note
☒	Ubuntu (Kernel 6.6)	Linux Driver	q324_Inxdri	q324_Inxdri	

Tool List

Update	Driver	Current Ver. in BIOS	Latest Ver. by Chipset Vendor	Note
	BIOS Test Suite	15.1.124.740	15.1.124.740	
	ApcbV3BinTool	1.0.0.12	1.0.0.12	
	RPMC	1.6.1.0	1.6.1.0	
	BIOS Build Tool: MSVC	14.28.293333(Visual Studio 2019)		Only in Windows and didn't support other compiler on Windows
	BIOS Build Tool: NASM	2.15.05		
	BIOS Build Tool: iASL	20230628		Newer version also can be used for BIOS building
	BIOS Build Tool: Python	3.11.3		Python

				3.10.x and 3.11.x also can be used for BIOS building. Python 2 can't support ed.
	BIOS Build Tool : Strawberry Perl	5.32.1.1		Optional . If want to re-generate CBS setup, it will need to use perl.
	Insyde H2OFFT	Shell 200.02.00.10		

Specification List

Update	Specification	Current Ver. in BIOS	Latest Ver. by Chipset Vendor	Note
	Processor Programming Reference (PPR) for AMD Family 19h Model 44h, Revision B1 Processors	3.07	3.07	
	AMD Platform Security Processor BIOS Architecture Design Guide for AMD Client Processors	2.02	2.02	
	AMD Generic Encapsulated Software Architecture (AGESA™) Interface Specification	1.72	1.72	

External SVN Path

https://code.insydesw.com/H2O-Embedded-CRB/AMD_V3000_Fp7MultiBoard_Rev5.4/Tag/05.46.50.0022

2. Release Note Key Message

Release Note Key Message Table

No.	Severity	Description	Link
1	Feature	(ITS:0128530) New Feature : default password (H2ODefaultPasswordDxe)	Detail
2	Feature	Update kernel to 05.46.44. For more details, please refer to Kernel.txt.	Detail
3	Feature	Update kernel to 05.46.50. For more details, please refer to Kernel.txt.	Detail
4	Feature	EmbeddedPI-FP7r2_100C AGESA - AGESA-V9 EmbeddedPI-FP7r2_100C for Family 19h Model 40h (Rembrandt) ABL (DDR5) - AGESA Boot Loader: RABLRM4C09704E ABL RT - AGESA Run Time Driver: RABLRTRM1C017011 ABL PMU - 16 AMF FW - 28.28.0.25	Detail

		DF RIB FW - 1.0.2.B DMCUB FW B0 - 4.0.15.47 MSMU FW - FF.45.2E.0 PSP FW - 0.28.A8.70 PSP fTPM - 3.77.0.5 DRTM - 4.11.0.2E Security Policy B0 - D.19.0.73 SMU FW B0 - 6.69.29.0 SPL Table - 5.28.0.6D SecPatchLevel - 5 UCODE B0 - 0A404002 B1 - 0A40410A USB FW - 0.1.0.15 VBL - 20.11.00.1C SpiRomConfig - 0.22.26.1 Prom-21 FW - 230215_A0_01_82 PD FW - 0310.00.5.3 AmdCbsPkg - Common BIOS Setup Options AmdCpmPkg - Common Platform Module v065 GOP - Rembrandt GOP driver signed v3.8.8 PlatformBIOS - Fox platform BIOS RFX100CB SmmSupervisorPkg - 3.011X VBIOS - 032 *Tools - AMDBIOSDBG: 2.06.14.01 - ApcbV3BinTool: 1.0.0.12 - BIOSTestSuite: 15.1.124.740 - PSP tool: 4.0.50 - RPMC: 1.6.1.0	
5	Serious	1. (ITS:0130114) AcpiS3SaveDxe/ChipsetSvcDxe: A potential DXE memory corruption vulnerability (CVE-2024-49200) 2. Use Lock Box to store 'AcpiVariableSet' pointer instead of using UEFI variable function to prevent memory corruption.	Detail
6	Trivial	Update BIOS revision to 05.46.50.0022	Detail

*** Description Clarification**

Silicon vendor reference code update
Project dependent item
Kernel / Common code update

*** Severity Clarification**

Severity	Clarification
Fatal	System hang
Serious	Function does not work
Medium	Partial function does not work, and it will not affect user normal usage
Minor	Code enhancement

Tweak	Code adjustment
Text	Error in the text, grammar or wording
Trivial	Trivial change
Feature	New feature, RC(AMD PI/AGESA and CIMX)update, kernel/Common platform/chipset update

3. Release Note

No.	Issue List	Related Files
1	Issue#: IB20840108 Severity: Feature Category: HDD Password Symptom: none RootCause: Missing H2ODefaultPasswordDxeString s.uni Description: (ITS:0128530) New Feature : default password (H2ODefaultPasswordDxe) Solution: none SolutionDepedency: IB20840100	Added: ----- 1. AMD\AmdEmbedded\EmbeddedChipsetPkg\ H2ODefaultPasswordDxe\ H2ODefaultPasswordDxeStrings.uni
2	Issue#: IB20840109 Severity: Feature Category: Code Sync Symptom: (Kernel update) RootCause: (Kernel update) Description: Update kernel to 05.46.44. For more details, please refer to Kernel.txt. Solution: (Kernel update) SolutionDepedency: none	Modified: ----- 1. AMD\AmdEmbedded\EmbeddedChipsetPkg\ MemoryIntiPei\CpuInstallEfiMemoryPei.inf 2. AMD\AmdEmbedded\EmbeddedChipsetPkg\ MemoryIntiPei\CrisisMemoryInitPei.inf 3. AMD\AmdEmbedded\EmbeddedChipsetPkg\ MemoryIntiPei\InstallPeiMemory.c 4. Board\AMD\FoxBoardPkg\BIOS\Fox.FD 5. Board\AMD\FoxBoardPkg\Project.dsc 6. Board\AMD\FoxBoardPkg\Project.uni 7. Kernel.txt
3	Issue#: IB20840117 Severity: Feature Category: Code Sync Symptom: (Kernel update)	Modified: ----- 1. Board\AMD\FoxBoardPkg\BIOS\Fox.FD 2. Board\AMD\FoxBoardPkg\Project.dsc 3. Board\AMD\FoxBoardPkg\Project.uni 4. Kernel.txt

	RootCause: (Kernel update) Description: Update kernel to 05.46.50. For more details, please refer to Kernel.txt. Solution: (Kernel update) SolutionDependency: none	
4	Issue#: IB20840120 Severity: Feature Category: Reference Code Update Symptom: (Reference Code Update) RootCause: (Reference Code Update) Description: EmbeddedPI-FP7r2_100C AGESA - AGESA-V9 EmbeddedPI-FP7r2_100C for Family 19h Model 40h (Rembrandt) ABL (DDR5) - AGESA Boot Loader: RABLRM4C09704E ABL RT - AGESA Run Time Driver: RABLRTRM1C017011 ABL PMU - 16 AMF FW - 28.28.0.25 DF RIB FW - 1.0.2.B DMCUB FW B0 - 4.0.15.47 MSMU FW - FF.45.2E.0 PSP FW - 0.28.A8.70 PSP fTPM - 3.77.0.5 DRTM - 4.11.0.2E Security Policy B0 - D.19.0.73 SMU FW B0 - 6.69.29.0 SPL Table - 5.28.0.6D SecPatchLevel - 5 UCODE B0 - 0A404002 B1 - 0A40410A USB FW - 0.1.0.15 VBL -	Added: ----- 1. AMD\AmdEmbedded\AgesaModulePkg\Firmwares\RMB\ 8024_rsmu_sec_policy.rmb_L0.0d190073.sbin 2. AMD\AmdEmbedded\AgesaModulePkg\Firmwares\RMB\ 8045_rsmu_sec_policy.rmb_L1.0d190173.sbin 3. AMD\AmdEmbedded\AgesaModulePkg\Firmwares\RMB\APCB.h 4. AMD\AmdEmbedded\AgesaModulePkg\Firmwares\RMB\APOB.h 5. AMD\AmdEmbedded\AmdCpmPkg\Features\AmdVariableProtection\AmdVariableProtectionLib.c 6. AMD\AmdEmbedded\AmdCpmPkg\Features\AmdVariableProtection\AmdVariableProtectionLib.inf 7. Board\AMD\FoxBoardPkg\ReleaseNotesPI_100C_RC2.txt Deleted: ----- 1. AMD\AmdEmbedded\AgesaModulePkg\Firmwares\RMB\ 8024_rsmu_sec_policy.rmb_L0.0d190072.sbin 2. AMD\AmdEmbedded\AgesaModulePkg\Firmwares\RMB\ 8045_rsmu_sec_policy.rmb_L1.0d190172.sbin 3. AMD\AmdEmbedded\AmdCpmPkg\Addendum\Oem\Fox\Dxe\NvmeRaidDxe\ 4. AMD\AmdEmbedded\AmdCpmPkg\Features\AmdVariableProtection\AmdVariableProtectionLibVarCheck.c 5. AMD\AmdEmbedded\AmdCpmPkg\Features\AmdVariableProtection\AmdVariableProtectionLibVarCheck.inf 6. AMD\AmdEmbedded\AmdCpmPkg\Features\AmdVariableProtection\AmdVariableProtectionSmm.c 7. AMD\AmdEmbedded\AmdCpmPkg\Features\AmdVariableProtection\AmdVariableProtectionSmm.inf 8. AMD\AmdEmbedded\AmdCpmPkg\Include\AmdVariableProtectionTable.h 9. Board\AMD\FoxBoardPkg\ReleaseNotesPI_100B_RC1.txt Modified: -----

20.11.00.1C		1. AMD\AmdEmbedded\AgesaModulePkg\
SpiRomConfig	-	AgesaFp7r2RmbModulePkg.inc.dsc
0.22.26.1		2. AMD\AmdEmbedded\AgesaModulePkg\
Prom-21 FW	-	AgesaModuleNbIoPkg.dec
230215_A0_01_82		3. AMD\AmdEmbedded\AgesaModulePkg\Fch\Keith\
PD FW	-	FchKeithCore\Keith\FchRmb.asi
0310.00.5.3		4. AMD\AmdEmbedded\AgesaModulePkg\Fch\Keith\
AmdCbsPkg	-	FchKeithCore\Keith\KTHwAcpi\
Common BIOS Setup Options		KTHwAcpiEnvService.c
AmdCpmPkg	-	5. AMD\AmdEmbedded\AgesaModulePkg\Firmwares\
Common Platform Module v065		RMB\AgesaBLReleaseNotes.txt
GOP	-	6. AMD\AmdEmbedded\AgesaModulePkg\Firmwares\
Rembrandt GOP driver signed		RMB\AppbDdrReleaseNotes.txt
v3.8.8		7. AMD\AmdEmbedded\AgesaModulePkg\Firmwares\
PlatformBIOS	- Fox	RMB\AppbDdrReleaseNotes_Quickboot.txt
platform BIOS RFX100CB		8. AMD\AmdEmbedded\AgesaModulePkg\Firmwares\
SmmSupervisorPkg	-	RMB\ReleaseNotes.txt
3.011X		9. AMD\AmdEmbedded\AgesaModulePkg\Firmwares\
VBIOS	- 032	RMB\ReleaseNotes_Ext.txt
*Tools	-	10. AMD\AmdEmbedded\AgesaModulePkg\
AMDBIOSDBG: 2.06.14.01		Firmwares\RMB\Rembrandt_FWs.xml
- ApcbV3BinTool: 1.0.0.12		11. AMD\AmdEmbedded\AgesaModulePkg\
- BIOSTestSuite:		Firmwares\RMB\SmuReleaseNotesRMB_B0.txt
15.1.124.740		12. AMD\AmdEmbedded\AgesaModulePkg\
- PSP tool: 4.0.50		Firmwares\RMB\
- RPMC: 1.6.1.0		TypeId0x01_PspBootLoader1_RMB.sbin
		13. AMD\AmdEmbedded\AgesaModulePkg\
		Firmwares\RMB\TypeId0x02_PspOS_RMB.sbin
		14. AMD\AmdEmbedded\AgesaModulePkg\
		Firmwares\RMB\TypeId0x0C_FtpmDrv_RMB.csbin
		15. AMD\AmdEmbedded\AgesaModulePkg\
		Firmwares\RMB\
		TypeId0x108_SmuFirmware_RMB_B0.sbin
		16. AMD\AmdEmbedded\AgesaModulePkg\
		Firmwares\RMB\
		TypeId0x112_SmuFirmware2_RMB_B0.sbin
		17. AMD\AmdEmbedded\AgesaModulePkg\
		Firmwares\RMB\
		TypeId0x13_PspEarlyUnlock_RMB.sbin
		18. AMD\AmdEmbedded\AgesaModulePkg\
		Firmwares\RMB\
		TypeId0x28_PspSystemDriver_RMB.sbin
		19. AMD\AmdEmbedded\AgesaModulePkg\
		Firmwares\RMB\
		TypeId0x30_AgesaBootloaderU_RMB_DDR5.sbin
		20. AMD\AmdEmbedded\AgesaModulePkg\
		Firmwares\RMB\TypeId0x55_SplTableBl_RMB.sbin
		21. AMD\AmdEmbedded\AgesaModulePkg\
		Firmwares\RMB\
		TypeId0x64_Appb_RMB_Ddr5QuickbootImem10.csbin
		22. AMD\AmdEmbedded\AgesaModulePkg\
		Firmwares\RMB\
		TypeId0x64_Appb_RMB_Ddr5QuickbootImem9.csbin
		23. AMD\AmdEmbedded\AgesaModulePkg\
		Firmwares\RMB\
		TypeId0x65_Appb_RMB_Ddr5QuickbootDmem10.csbin
		24. AMD\AmdEmbedded\AgesaModulePkg\
		Firmwares\RMB\
		TypeId0x65_Appb_RMB_Ddr5QuickbootDmem9.csbin
		25. AMD\AmdEmbedded\AgesaModulePkg\

Solution:

(Reference Code Update)

SolutionDependency: none

	Firmwares\RMB\ TypeId0x66_UcodePatch_RMB_B1.sbin 26. AMD\AmdEmbedded\AgesaModulePkg\ Firmwares\RMB\ TypeId0x73_PspBootLoader2_RMB.sbin 27. AMD\AmdEmbedded\AgesaModulePkg\Include\ AmdPspSmmCommunication.h 28. AMD\AmdEmbedded\AgesaModulePkg\Include\ IdsNvDefRMB.h 29. AMD\AmdEmbedded\AgesaModulePkg\Include\ IdsNvIdRMB.h 30. AMD\AmdEmbedded\AgesaModulePkg\Library\ AmdPspFlashAccLibDxe\AmdPspFlashAccLibDxe.c 31. AMD\AmdEmbedded\AgesaModulePkg\Nbio\SMU\ RMB\DXE\InfoCollection.c 32. AMD\AmdEmbedded\AgesaModulePkg\Nbio\SMU\ RMB\DXE\SmuV13DxeRMB.inf 33. AMD\AmdEmbedded\AgesaModulePkg\ Package.dsc 34. AMD\AmdEmbedded\AgesaModulePkg\Psp\ AmdPspP2CmboxV2\AmdPspFlashAccSmmHdlr.c 35. AMD\AmdEmbedded\AgesaModulePkg\Psp\ AmdPspP2CmboxV2\AmdPspP2CmboxV2.h 36. AMD\AmdEmbedded\AgesaModulePkg\Psp\ AmdPspRomArmor2Smm\AmdPspRomArmor2Smm.c 37. AMD\AmdEmbedded\AgesaModulePkg\Psp\ AmdPspRomArmor3Smm\AmdPspRomArmor3Smm.c 38. AMD\AmdEmbedded\AgesaPkg\Addendum\Apcb\ Inc\RMB\ApcbV3TokenUid.h 39. AMD\AmdEmbedded\AgesaPkg\Addendum\Apcb\ RembrandtFp7r2\Include\ApcbDefaults.h 40. AMD\AmdEmbedded\AmdCbsPkg\Build\ IdsNvIdRMB.h 41. AMD\AmdEmbedded\AmdCbsPkg\Build\ ResourceRMB\AmdCbsDefault.c 42. AMD\AmdEmbedded\AmdCbsPkg\Build\ ResourceRMB\AmdCbsForm.vfr 43. AMD\AmdEmbedded\AmdCbsPkg\Build\ ResourceRMB\AmdCbsStrings.uni 44. AMD\AmdEmbedded\AmdCbsPkg\Build\ ResourceRMB\AmdCbsVariable.h 45. AMD\AmdEmbedded\AmdCbsPkg\Build\ ResourceRMB\ApcbSetDataRMB.c 46. AMD\AmdEmbedded\AmdCbsPkg\Build\ ResourceRMB\CbsIdsNvRMB.c 47. AMD\AmdEmbedded\AmdCbsPkg\Build\ ResourceRMB\CbsSetAgesaPcdLibRMB.c 48. AMD\AmdEmbedded\AmdCbsPkg\Build\ ResourceRMB\CbsSetAgesaPcdLibRMB.inf 49. AMD\AmdEmbedded\AmdCbsPkg\Build\ ResourceRMB\IdsNvDefRMB.h 50. AMD\AmdEmbedded\AmdCbsPkg\Build\ ResourceRMB\IdsSampleRMB.c 51. AMD\AmdEmbedded\AmdCbsPkg\Library\ Family\0x19\RMB\External\Rembrandt\ eSetupFox.xml 52. AMD\AmdEmbedded\AmdCpmPkg\Addendum\Oem\ Fox\AmdCpmPkg.dxe.inc.fdf 53. AMD\AmdEmbedded\AmdCpmPkg\Addendum\Oem\ Fox\AmdCpmPkg.inc.dsc 54. AMD\AmdEmbedded\AmdCpmPkg\Addendum\Oem\ Fox\Dxe\AmdPbsSetupDxe\AmdPbsHii.uni
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		55. AMD\AmdEmbedded\AmdCpmPkg\Addendum\Oem\Fox\Dxe\AmdPbsSetupDxe\AmdPbsHii.vfr 56. AMD\AmdEmbedded\AmdCpmPkg\AmdCpmPkg.dec 57. AMD\AmdEmbedded\AmdCpmPkg\Features\AmdVariableProtection\AmdVariableProtection.c 58. AMD\AmdEmbedded\AmdCpmPkg\Features\AmdVariableProtection\AmdVariableProtection.inf 59. AMD\AmdEmbedded\AmdCpmPkg\Features\AmdVariableProtection\ReadMe.txt 60. AMD\AmdEmbedded\AmdCpmPkg\Features\DisplayFeature\Smm\AmdCpm3DisplayFeatureSmm.inf 61. AMD\AmdEmbedded\AmdCpmPkg\Features\DisplayFeature\Smm\AmdCpm3DisplayFeatureSmmBr.inf 62. AMD\AmdEmbedded\AmdCpmPkg\Features\DisplayFeature\Smm\DisplayFeatureSmiRegistration.c 63. AMD\AmdEmbedded\AmdCpmPkg\Features\GpioInit\Smm\AmdCpmGpioInitSmm.c 64. AMD\AmdEmbedded\AmdCpmPkg\Include\AmdCpmCommon.h 65. AMD\AmdEmbedded\AmdCpmPkg\Include\AmdCpmDefine.h 66. AMD\AmdEmbedded\AmdCpmPkg\Include\AmdCpmFunction.h 67. AMD\AmdEmbedded\AmdCpmPkg\Include\AmdVariableProtectionItem.h 68. AMD\AmdEmbedded\AmdCpmPkg\Include\Library\AmdVariableProtectionLib.h 69. AMD\AmdEmbedded\AmdCpmPkg\Kernel\Dxe\AmdCpmInitDxe.c 70. AMD\AmdEmbedded\AmdCpmPkg\Kernel\Smm\AmdCpm3InitSmm.inf 71. AMD\AmdEmbedded\AmdCpmPkg\Kernel\Smm\AmdCpmInitSmm.c 72. AMD\AmdEmbedded\AmdCpmPkg\Library\Protocol\AmdCpmTableProtocol\AmdCpmTableProtocol.h 73. AMD\AmdEmbedded\AmdCpmPkg\Package.dsc 74. AMD\AmdEmbedded\AmdCpmPkg\Package.fdf 75. AMD\AmdEmbedded\EmbeddedChipsetPkg\SmmAccessPei\SmmAccessPeim.c 76. AMD\AmdEmbedded\EmbeddedChipsetPkg\SnpDxe\AmdXgbe.c 77. Board\AMD\FoxBoardPkg\ApcbV3Customization\ApcbCustomizedDefinitionsRmbFp7DDR5.h 78. Board\AMD\FoxBoardPkg\BIOS\Fox.FD 79. Board\AMD\FoxBoardPkg\BIOS\ResizeRegion.txt 80. Board\AMD\FoxBoardPkg\PlatformConfig\BIOSImageDirectory32M_DDR5.xml 81. Board\AMD\FoxBoardPkg\Project.dsc 82. Board\AMD\FoxBoardPkg\Project.uni 83. Board\AMD\FoxBoardPkg\ProjectSourceCodeAnalysis.csv
5	Issue#: IB20840114	Modified: -----

	Severity: Serious Category: Security Symptom: (Security) RootCause: (Security) Description: 1. (ITS:0130114) AcpiS3SaveDxe/ChipsetSvcDxe : A potential DXE memory corruption vulnerability (CVE-2024-49200) 2. Use Lock Box to store 'AcpiVariableSet' pointer instead of using UEFI variable function to prevent memory corruption. Solution: (Security) SolutionDependency: IB20840109	1. AMD\AmdEmbedded\EmbeddedChipsetPkg\Compatibility\ AcpiVariableHobOnSmramReserveHobThunk\ AcpiVariableHobOnSmramReserveHobThunk.c 2. AMD\AmdEmbedded\EmbeddedChipsetPkg\ SmmAccessPei\SmmAccessPei.inf 3. AMD\AmdEmbedded\EmbeddedChipsetPkg\ SmmAccessPei\SmmAccessPeim.c 4. AMD\AmdEmbedded\EmbeddedChipsetPkg\ SmmAccessPei\SmmAccessPeim.h 5. Board\AMD\FoxBoardPkg\BIOS\Fox.FD
6	Issue#: IB20840000 Severity: Trivial Category: Tag Symptom: (Revision Update) RootCause: (Revision Update) Description: Update BIOS revision to 05.46.50.0022 Solution: (Revision Update) SolutionDependency: none	